

Math check: debunking “AI wastes water” with AI vs human water use per artwork (two scopes, citations)

Posted by u/PrinceLucipurr • 0 points • 7 comments

TL;DR (ratios + caveats)

Scope A (operational only)

0.53 : 1 (AI : Human) for a typical ~4h comparable artwork.

- **Swings with time + grid**
- **~1h** sketch (often low detail, likely no colour): AI can be higher
- **~10h** high detail: AI is typically much lower

Scope B (adds human marginal food water)

1 : 3,500 (AI : Human) for the same ~4h comparable artwork.

- **3,500 is NOT drinking water**
- It is **water used to produce the food** for extra calories burned
- Varies massively with **diet** and **time**

If you want the full breakdown, I posted the longer version and citations in the comments.

Comments

u/pikapika200 • 3 points • 2026-01-24 17:46 UTC

does that girl's pendant have magical powers?

u/PrinceLucipurr • 2 points • 2026-01-24 17:53 UTC

Yes Pikapika200 it does indeed!

It allows the use of HM01 (Cut).

It ensures that all Pokémon (including those obtained through trades) up to Level 30 will obey your commands in battle. Traded Pokémon above this level may refuse to attack or loaf around.

<https://i.redd.it/2479qx2h5cfg1.gif>

u/pikapika200 • 2 points • 2026-01-24 18:48 UTC

wait, why is she wearing the Cascade Badge as a pendant?

u/PrinceLucipurr • 1 points • 2026-01-24 18:55 UTC

That was just me making a pun, after I saw your name 🐱 It just meant to be a water drop pendant, as I felt it was a nice aesthetic for the genre of the video and post 🐱

u/PrinceLucipurr • 2 points • 2026-01-24 14:37 UTC

Full breakdown (two scopes, with the key numbers)

Scope A: Operational only (direct, attributable)

AI water per image = $(\text{kWh per image}) \times (\text{litres per kWh})$

Typical ranges:

- AI can be anywhere from $\sim 0.005 \text{ L}$ per image (very efficient + low water intensity grid)
- Around $\sim 0.225 \text{ L}$ per image (typical setup)
- Up to $\sim 8+ \text{ L}$ per image (high energy + water intensive grid)

Human drinking water over work time (roughly $\sim 0.11 \text{ L per hour}$):

- 1 h : $\sim 0.11 \text{ L}$ (often low detail, likely no colour)
- 4 h : $\sim 0.43 \text{ L}$
- 10 h : $\sim 1.07 \text{ L}$

So the “AI always wastes bottles” claim is false. It depends on model efficiency, grid water intensity, and how long the human equivalent actually takes.

Scope B: Add marginal food water (still finite, no infinite regress)

Humans run on food. Extra calories burned while working have embedded water costs.

Even 1 hour of drawing can represent tens to thousands of litres of water depending on diet.

That’s why, once food is included, human side typically dominates massively.

u/PrinceLucipurr • 1 points • 2026-01-24 14:38 UTC

AI “wastes bottles of water” per image? Let’s do the maths (two scopes)

Scope A: Operational only (direct, attributable)

Rule: only count direct consumption caused by making the image.

- **AI:** electricity per image converted to water (litres per kWh)
- **Human:** drinking water needed over the hours spent drawing

Formula

- AI water per image (L) = $(\text{kWh per image}) \times (\text{L per kWh})$

Reasonable ranges used

AI energy per image:

- Low: 0.003 kWh
- Typical: 0.05 kWh
- High: 0.12 kWh

Water intensity of electricity:

- Low: **1.8 L/kWh**
- Typical: **4.5 L/kWh**
- High: **68 L/kWh**

AI results (water per image)

- Low + low: **0.0054 L** (about a teaspoon)
- Typical + typical: **0.225 L** (about a glass)
- High + high: **8.16 L** (multiple litres)

Human drinking water (time-shifted still counts)

Guideline total fluids per day: ~**2.7 L** (women) to ~**3.7 L** (men). About **80%** is beverages.

So beverages per hour roughly:

- **0.09 to 0.13 L/hour** (typical ~**0.11 L/h**)

If the same piece takes:

- **1 hour:** ~**0.11 L** (often a low detail sketch, likely no colour)
- **4 hours:** ~**0.43 L**
- **10 hours:** ~**1.07 L**

Head-to-head examples (typical AI at 0.225 L)

- vs **1h** sketch (~0.11 L): AI about **2× higher**
- vs **4h** illustration (~0.43 L): AI about **half**
- vs **10h** high detail (~1.07 L): AI about **one fifth**

So the “AI is always wasting bottles” take is false. It depends on model efficiency, grid water intensity, and how long the comparable human work takes.

Scope B: Add marginal food water (still finite)

Rule: only add **extra calories burned during the work session**, not “your entire life cost” or agriculture infrastructure.

Calories burned while drawing: ~**125 to 175 kcal/hour**

Water footprint per calorie (diet dependent):

- Grains: **0.51 L per calorie**
- Vegetables: **1.34 L per calorie**
- Fruit: **2.09 L per calorie**
- Beef: **10.19 L per calorie**

Marginal food water for 1 hour drawing (125 to 175 kcal)

- Grains: ~**64 to 89 L**
- Vegetables: ~**168 to 235 L**
- Fruit: ~**261 to 366 L**
- Beef-heavy: ~**1,274 to 1,783 L**

For **4 hours**, multiply by 4. For **10 hours**, multiply by 10.

This typically dwarfs AI operational water.

****Bottom line****

- If someone is whining about “bottles of water per image”, use ****Scope A****.
- If they widen it to “energy”, humans run on food, and food has a massive embedded water footprint.
- If they try to drag it into manufacturing or “humans were raised by civilisation”, that’s infinite regress, not a serious comparison.

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Evidence pack (sources)

1) AI energy per image (kWh)

- Luccioni, Jernite, Strubel (arXiv): “Power Hungry Processing: Watts Driving the Cost of AI Deployment”

<https://arxiv.org/abs/2311.16863>

Supports: energy per 1,000 inferences (includes image generation), convertible to kWh per image.

2) Water intensity of electricity (L per kWh)

- NREL (Torcellini et al.): “Consumptive Water Use for U.S. Power Production”

<https://docs.nrel.gov/docs/fy04osti/33905.pdf>

Supports: water consumption per kWh ranges (thermoelectric vs hydro, plus national average).

3) Data centres and indirect water via electricity

- EESI: “Data Centers and Water Consumption”

<https://www.eesi.org/articles/view/data-centers-and-water-consumption>

Supports: indirect water use estimates tied to electricity generation.

4) Human daily fluid intake (baseline hydration)

- National Academies (DRI): Water intake chapter

<https://www.nationalacademies.org/read/10925/chapter/6>

Supports: ~3.7 L/day (men) and ~2.7 L/day (women), and that most comes from beverages.

5) Food water footprint (for Scope B)

- Water Footprint Network: Mekonnen and Hoekstra (2012): “A Global Assessment of the Water Footprint of Farm Animal Products”

[https://www.waterfootprint.org/resources/Mekonnen-Hoekstra-2012-](https://www.waterfootprint.org/resources/Mekonnen-Hoekstra-2012-WaterFootprintFarmAnimalProducts_1.pdf)

[WaterFootprintFarmAnimalProducts_1.pdf](https://www.waterfootprint.org/resources/Mekonnen-Hoekstra-2012-WaterFootprintFarmAnimalProducts_1.pdf)

Supports: very high water footprint for animal products, especially beef.